This turns the sinc function into $x/x$, which has a value of *one*. In other words, as $x$ becomes smaller and smaller, the value of $sinc(x)$ approaches *one*, which includes $sinc(0) = 1$. Now try to tell your computer this! All it sees is a division by zero, causing it to complain and stop your program. The important point to remember is that your program must include special handling at $x = 0$ when calculating the sinc function.

A key trait of the sinc function is the location of the **zero crossings**. These occur at frequencies where an integer number of the sinusoid's cycles fit evenly into the rectangular pulse. For example, if the rectangular pulse is 20 points wide, the first zero in the frequency domain is at the frequency that makes one complete cycle in 20 points. The second zero is at the frequency that makes two complete cycles in 20 points, etc. This can be understood by remembering how the DFT is calculated by correlation. The amplitude of a frequency component is found by multiplying the time domain signal by a sinusoid and adding up the resulting samples. If the time domain waveform is a rectangular pulse of unity amplitude, this is the same as *adding* the sinusoid's samples that are within the rectangular pulse. If this summation occurs over an integral number of the sinusoid's cycles, the result will be zero.

The sinc function is widely used in DSP because it is the Fourier transform pair of a very simple waveform, the rectangular pulse. For example, the sinc function is used in *spectral analysis*, as discussed in Chapter 9. Consider the analysis of an infinitely long discrete signal. Since the DFT can only work with *finite* length signals, $N$ samples are selected to represent the longer signal. The key here is that "selecting $N$ samples from a longer signal" is the same as multiplying the longer signal by a rectangular pulse. The *ones* in the rectangular pulse retain the corresponding samples, while the *zeros* eliminate them. How does this affect the frequency spectrum of the signal? Multiplying the time domain by a rectangular pulse results in the frequency domain being *convolved* with a *sinc function*. This reduces the frequency spectrum's resolution, as previously shown in Fig. 9-5a.

## Other Transform Pairs

Figure 11-5 (a) and (b) show the duality of the above: a rectangular pulse in the frequency domain corresponds to a sinc function (plus aliasing) in the time domain. Including the effects of aliasing, the time domain signal is given by:

EQUATION 11-3
Inverse DFT of the rectangular pulse. In the frequency domain, the pulse has an amplitude of one, and runs from sample number 0 through sample number $M$-1. The parameter $N$ is the length of the DFT, and $x[i]$ is the time domain signal with $i$ running from 0 to $N$-1. To avoid the division by zero, use $x[0] = (2M-1)/N$.

$$x[i] = \frac{1}{N} \frac{\sin(2\pi i (M - 1/2)/N)}{\sin(\pi i / N)}$$